Technical report — Gegenprobe: one schema and a converter for Swiss parliamentary answers

Track 2A — OpenParlData · Online · Team 444 — Roman (+ Claude Code agents, §7) · Evidence site https://gegenprobe.mi11ion.io · **Maturity:** census and Bern repair lists = *ready to apply* (relinks: one OPD decision, its hash rule); source-adapter lists = *audited*; nightly job = *deployable, two real nights*; rules-only converter = *tested prototype*; review mode, whole and virtual affairs = *feasibility study*; schema v0.1 = *draft for the eCH group*.

1. Summary

The answer. A common schema for the business of Swiss parliaments works if it carries roles. "Answer date" is the government's decision in some parliaments, the letter or registration date in others and the submission date in a few; schema v0.1 gives every date its role, every value its source (page and character span, or the OpenParlData (OPD) record) and every empty field a typed reason.

1. **Schema.** Five fields are measured to need a role split (answer date, procedural history, type, procedural state, decision outcome), and v0.1 encodes each one. In 7 parliaments OPD's "answer" event is mostly the submission date; median answer times are 112 days where it is the verified decision date, 79 on a government's weekly sitting and 39 for diffuse, publication-like dates. The other 12 fields are defined once in v0.1; a field-by-field meaning test is the next step with the eCH group, set out as P8 for seven of them (`eval/prereg9/`; Table 1).
2. **Census.** Of 91,268 interventions with a written government answer, **92.3%** have answer text once the counted repairs are applied (82.3% today). At most 13,150 written answers lack text on their intervention; 2,177 of the "missing" answers were in OPD all along under a heading the census lexicon does not list; in Bern a linking bug, confirmed by OPD (correspondence, 10 Oct 2026), leaves most answers unlinked.
3. **Ready for OPD.** Repairs proposed for **9,720** answers, keyed on OPD ids: Bern 2,421 relinks, 48 fetches and 554 `begin_date` corrections with a dry-run SQL plan (and a guarded plan for the tag rows); per-parliament lists from 50 model-free source adapters (blind audit 197/200 are the executive's answer); 9 OPD vocabulary labels wrong as published (e.g. "Votaziun" on 9,887 affairs); a Fedlex resolver for 29,946 federal documents that store Fedlex's "JavaScript required" page. Schema v0.1 also ships as one standalone file for OPD's workbench (`schema/gegenprobe_schema.py`, pydantic only).
4. **Apertus.** It picks the right printed date where the rules cannot: as a closed-set verifier (P7) it recovers **214** review-queue answers, 120/120 sampled promotions are the government's answer, and its choice of date lifts date-checked precision by 20.9 points [5.2, 33.1] over the adapters' date rule (post-hoc ablation). In parliaments nobody wrote rules for it finds 24/29 answer dates (rules 2/29, P2), and it doubles submitter extraction (P5: 12/40 → 24/40). Every value is a date or phrase printed in the document, with page and span: free-prompt accuracy, with provenance guaranteed by construction.

2. Architecture

Census (`census/`) streams the OPD exports once into a per-affair *answer-link status* (`attached | agenda | parentless | absent | image_only | oral`), the feasibility matrix and a comparability measure; a derived layer (`census/corrected.py`) places every registered gap row once the repair routes have looked (Fig. 1). **Repair** (`repair/`): deterministic routes per cause, each row checked (affair number or title, answer heading, date against OPD's answer event), as lists keyed on OPD ids; a nightly job (`deploy/nightly/`) re-runs the Bern route. **Converter** (`convert/`, `qa/`): OPD metadata and events, then frozen DE/FR/IT anchor rules, then the Apertus closed-set pointer on the residue; modes `rules_only` (default, no model) and `review` (suggestions with page, span and margin, flagged `needs_review`). **Schema and validator** (`schema/`, `validate/`): Pydantic schema v0.1, normative; a value carries {`value, role, page, char_start, char_end, quote, route, rule, model_build, θ, margin`}, a null its reason (`not_in_document | unlinked | image_only | oral | text_failed | present_unanchored | no_rules_for_body | not_applicable`); `QAItem` pairs question *n* with its answer passage.

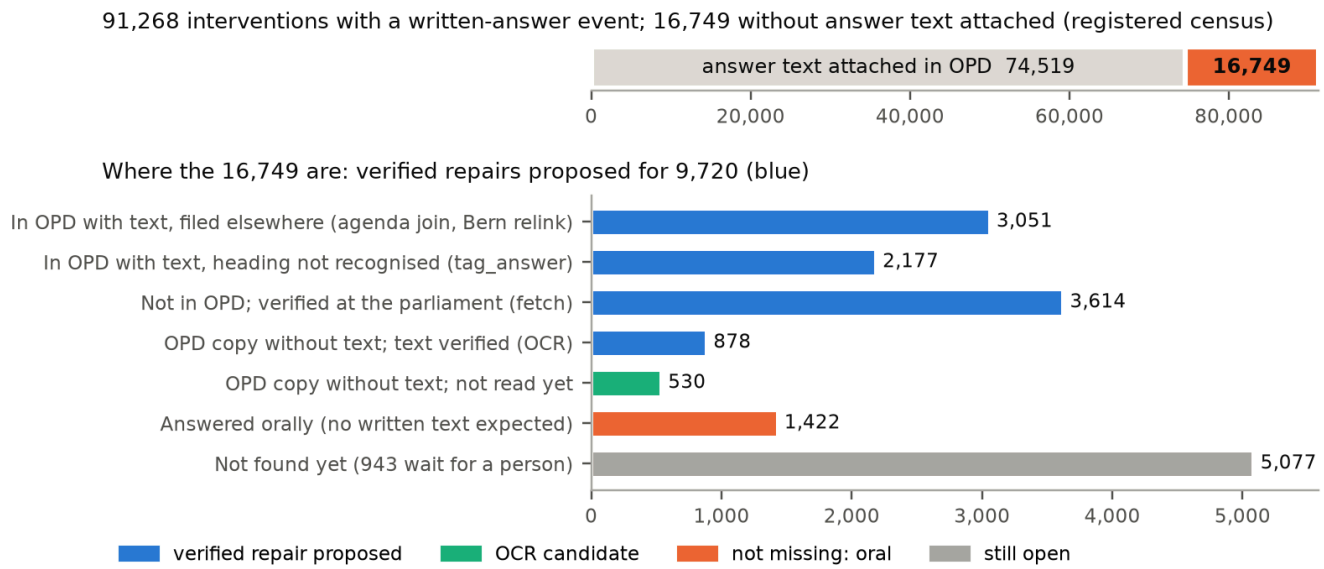


Figure 1. The registered 16,749 placed once each (`results/census/corrected.json`). Still to find: 5,077 (943 of them already waiting for a person).

3. Use of Apertus

Closed form only. Apertus never writes a value. (i) *Verifier* (P7, `repair/adapters/adjudicate.py`): for an adapter review row it answers "does this document print the executive's answer to this intervention (number, title)?" (yes vs no scored after a JSON prefix, θ_A = 8) and picks the answer date among the printed dates or none (θ_B = 10); a row is promoted only if both pass and the date passes the adapters' own date rule. (ii) *Pointer* (`convert/pointer.py`, prompts `src/prompts/{de,fr,it}`): the document (first 12,000 characters) and the question in the user message, the assistant turn prefilled with a JSON prefix, `null` and every distinct printed date scored by summed log-probability; null if `lp(null) − logsumexp(lp(dates)) > θ`. Code locates every span (`pointer.locate`: the occurrence nearest an answer anchor, else the first). θ does not transfer across builds, so it is fitted per build on dev documents and frozen (θ = 10.40, AUROC 0.99). The same form picks labels and, in `qa/`, the question a paragraph answers, the recommendation sentence or the submitter's printed name. **Why closed:** asked freely, Apertus dates 15/15 (8B) and 14/15 (70B) Bern interventions that have no answer. **Design:** Apertus suggestions go to a reviewer; rules run unattended.

Which model. Evaluation used **Apertus v1.5 70B weights served by Public AI** under the name `swiss-ai/apertus-v1.5-8b` (inferred from the served fingerprint `tp8-pp2-…` and 760/792 greedy outputs byte-identical to the `-70b` name); the response `model` is checked on every call. The **8B** (`swiss-ai/Apertus-v1.5-8B`, a community 4-bit GGUF, sha256-pinned) runs locally on CPU with its own θ = 16.73 (exploratory): on P2's unseen parliaments it finds 15 of 21 answer dates (70B 16), and the registered P2-H2 test over the rules gives 13 vs 0; its two-stage reader v2 cuts dates on answerless documents from 18/34 to 5/34 (P6; v2 51/67 vs v1 48/67; rules then v2 55/67 vs rules 54/67). **Recommendation:** `rules_only` unattended; for suggestions the 8B where no text may leave the machine, the 70B for accuracy on infrastructure the operator chooses. Review mode stops (exit 3) on a served build θ was not fitted on.

Cost (`results/report/cost_end_to_end.json`, one core, no GPU). Fetch, extraction and checks about 2.1 s per document plus 6.9 s per OCR page; the rules process 480 documents per second (0.5 CPU-hours for all 902,754 documents in OPD's export). Review mode: 3.5 calls and 5,905 billed prompt tokens per residue document on the 70B, USD 0.59 per 1,000 residue documents; P7's verifier on the whole queue cost USD 0.50. The 8B takes 180 s per document on 4 vCPUs.

4. Data and operations

Data. OPD bulk exports (snapshot 6 Oct 2026; CC BY 4.0, "Source: OpenParlData.ch"; OPD's API robots.txt disallows `/v1/`), Bern open data, Fedlex, parliament sites, fetched politely (≥ 1.2 s per host and process). `data/` and `results/` are CDLA-Permissive-2.0 and carry OPD's attribution. **Sovereignty.** Runtime external services: none by default (OPD's exports and the parliaments' own sites); an optional review-model endpoint is chosen by the operator (Public AI, or the local 8B on CPU). **Operations.** `make run` and the nightly unit run the image read-only, as a non-root uid, with `--cap-drop ALL`; PDFs are capped at 50 MB and 600 s. One real night under systemd (83 min, Bern list reproduced on live data), then **two consecutive real nights in the recommended Docker setup** (Actions run 38098822336): night 1 took 86 min and streamed 6.4 GB; night 2 took 6.6 min (12 bodies re-scanned), 0 alerts; the ledger-status bug that night 2 exposed is fixed and covered by a two-night test.

Ready to apply: the dry-run plans are keyed on OPD ids; the only open decision is OPD's hash rule for relinks. **robots.txt:** we found, audited and fixed our own robots.txt bug; every fetch path now fails closed (disclosure: `docs/incidents.md`).

5. Results without a model: schema verdict, census, repair

Table 1. The 17 fields. *Class*: does the field mean the same thing in every parliament (`results/census/comparability.json`)? *Where it lives*: parliaments where OPD metadata carries it · carries it for part of the affairs · only the PDF · not in OPD's data (`results/census/matrix.json`).

field	class	evidence	where it lives (metadata · mixed · PDF only · none)
answer date	role split	decided, dispatched or registered; in 7 of 54 bodies mostly the submission date	**metadata**: 53 · 1 · 0 · 0
procedural history	role split	35 event labels carry different date roles	**metadata**: 78 · 9 · 1 · 1
type	role split	42 of 1,038 raw labels map to different types	partly: 61 · 24 · 2 · 2
procedural state	role split	32 labels ambiguous	partly: 41 · 27 · 18 · 3
decision outcome	role split	3 labels ambiguous	partly: 7 · 31 · 33 · 6
government recommendation	shared	OPD's labels map unambiguously (string check)	partly: 3 · 8 · 56 · 10
submission date	shared	defined once in v0.1	**metadata**: 89 · 0 · 0 · 0
answer-link status	shared	defined once in v0.1	**metadata**: 54 · 0 · 0 · 0
submitter	shared	defined once in v0.1	partly: 23 · 61 · 5 · 0
addressed body	shared	defined once in v0.1	partly: 12 · 10 · 54 · 10
decision date	shared	defined once in v0.1	partly: 7 · 30 · 34 · 6
Q/A pairing	shared	defined once in v0.1	partly: 22 · 25 · 4 · 3
language	shared	defined once in v0.1	partly: 14 · 53 · 20 · 2
usable text	shared	defined once in v0.1	partly: 41 · 42 · 0 · 6
co-signatories	shared	defined once in v0.1	PDF only: 0 · 2 · 72 · 12
dispatch date	shared	defined once in v0.1	PDF only: 0 · 0 · 47 · 7
related affairs	shared	defined once in v0.1	PDF only: 0 · 0 · 76 · 13

4 fields are in OPD's metadata almost everywhere, 13 partly or only in the PDF. The recommendation's string check (0 of 183 raw labels map to two concepts) covers the labels OPD has harmonised (29.1% of all events carry no harmonised id). Three further *concepts* are body-specific: oral answers in question time (23.6% of question-time answers), executive business that answers no member's affair, and councils that publish only meeting documents (16 bodies; v0.1 marks affairs rebuilt from minutes `virtual`, P4).

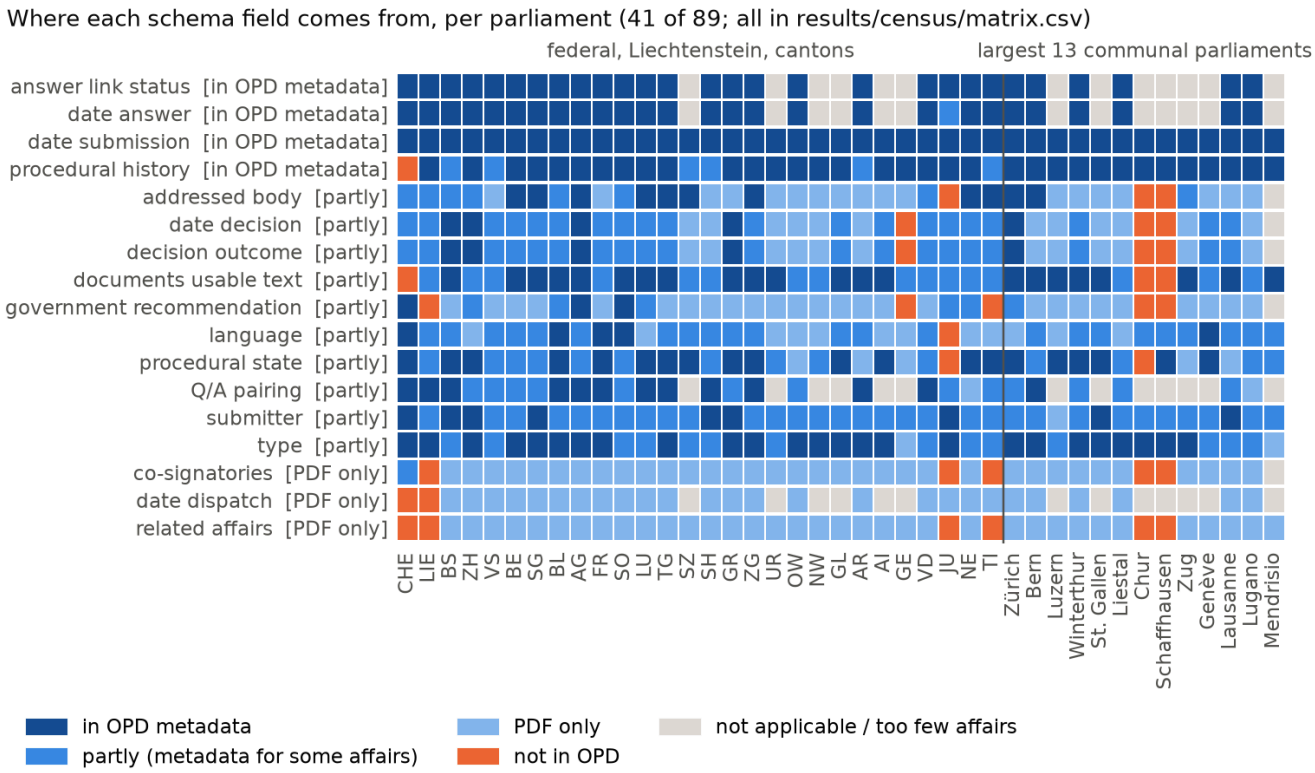


Figure 2. Where each field comes from, per parliament. Orange = not in OPD's data for that parliament (the parliament may still publish it).

Cross-parliamentary question. Of the 91,268 interventions with a written government answer, 81.6% have answer text attached in OPD (the registered count), 82.3% counting answers filed under an agenda item of the affair, 85.0% after the Bern relink and 92.3% after every counted repair. Answer times are not one quantity: the median is 112 days where OPD's "answer" date is the verified decision date (Bern, Ticino), 79 on a government's weekly sitting, 39 for diffuse, publication-like dates; in 7 parliaments the "answer" is the submission date.

Repair. *Bern*: of 3,532 answers the canton publishes OPD attaches 229 (6.5%), although 96.1% are in OPD and linkable without a model. 2,459 relink candidates = 2,421 that pass every check against OPD's own copy (affair number, heading, decision date = OPD's answer event [97.9, 98.9]) + 38 to review. The document `hash` is sha256(body_key + parent_type + url) (49,810 of 49,810 rows), so relinking changes each identifier and file URL unless OPD keeps the old one; the plan carries both values. *Source adapters* (`docs/adapters.md`): readers for 50 parliaments (27 yield counted rows) verify 6,613 answers by affair number or title, the parliament's answer formula and a date check, in three tiers (own date equals an OPD event 5,161; a printed date equals an OPD answer event 1,056; documented window 396). 2,177 of them are already in OPD with text under a heading the census lexicon does not list: the repair is a category on the document (`tag_answer`). Listed separately: 282 Jura answers matched by identity and wording only, and 917 review rows (214 of them recovered by Apertus into the tier `promoted_by_apertus`, §6). **Blind audit v3** (`results/repair/adapters/audit3/`): 200 counted rows and 60 decoys, shuffled into packets without the pipeline's verdict; the key's hash reached GitHub before the Claude verdicts were finished. Claude Opus judged 197/200 counted rows to be the executive's answer (Wilson 95.7% to 99.5%), and the date wrong or unsupported on 15. It rejected 60% of the failed decoys but only 1.5% of the counted rows; a second auditor from another vendor (Qwen-SEA-LION) ran on the same packets.

6. Evaluation: eight pre-registrations

Design. Each test froze code, prompts, θ and the scorer by sha256 before any label was committed. Unit (P1, P2, P6): one document's answer date, strict. Arms on identical inputs: **rules**; **pointer**; **cascade** (rules, then the pointer on the residue); **naive** free generation, plain and null-allowed; trivial baselines. **Gold, no human labelling by design:** two blind agent annotators (Claude Opus, Sonnet), a third breaks ties, quotes verified by code, raw files committed; all 84 annotator sessions ended before GitHub received their labels (`results/eval/annotation_sessions.json`). P1's rules were committed with its sample (D12), so P2 is the clean out-of-sample test of the rules.

Table 2. The eight tests, in registration order.

test	n	what it tests	registered result	note
P1	154 documents, parliaments with rules	answer date: rules vs Apertus	**partial**: H3 held; H1 null safety failed (7/90); H2 Branch B	rules alone 135/154, so no model by default
P2	62 documents, 15 parliaments without rules	rules transfer; pointer vs rules	**3 of 4 held**; H3 null safety failed (4/33)	pointer 24/29 dates, rules 2/29
P3	29 whole affairs, 10 parliaments	7 fields: metadata, rules, Apertus	**partial**: no field unattended; Apertus recall held, precision below	feasibility study (D3-5)
P3b	36 fresh affairs, 9 parliaments	replication of P3	**held**: replicated (S1, S2)	S2 lenient; type 35/36; addressed body 2/33
P4	120 agenda items, 6 sittings	affairs rebuilt from minutes	**partial**: H2, H4 held; H1, H3 failed	recall 99/120, precision 99/99
P5	40 fresh affairs, 10 parliaments	Q/A pairs, people, recommendation	**partial** tier (H1); H2, H5 held; H4 as rules vs free generation; H3 no gain	*not yet* when parliaments are resampled
P6	68 fresh documents, 8 parliaments	answer-date reader v2 vs free prompting	**no hypothesis held**	free prompt 62/68, v2 59/68
P7	478 review rows, 190 blind packets	Apertus as verifier of the review queue	**H1 holds** (120/120); H4 supported; H3 not supported	214 of 917 rows promoted

Unattended (95% lower bound ≥ 0.80, document-level intervals): the rules' answer date where rules exist (P1, helped by right nulls: dates found 47/64), the type on whole affairs (P3b, 35/36), and in P5 question detection (recall 167/187, precision 167/175), the reasoning span (37/40) and co-signatory precision (101/103). **Cross-checks.** `Qwen-SEA-LION-v4-32B-IT` (AI Singapore) relabelled P1, P2, P6 and P7 blind: date κ 0.80, 0.87 and 0.86, P7's is-answer labels 185/190; no registered decision changes. On P5 its labels pass the quote check in 7/40 files, so the recommendation count rests on the Claude gold. Non-LLM checks: OPD's event matches 37 of 44 checkable P1 dates; P6: 10/10, every gold quote on its page (34/34). Holm over all 20 registered confirmatory tests of P1–P7: the 9 that held survive (`results/eval/posthoc_r8_multiplicity.json`); whole parliaments are resampled as well.

P1 arm (154 documents)	exact [95% CI]	answered: found /64	wrong printed date	invented	answerless dated /90
rules only (default)	**135/154 [81.5, 92]**	47	4	0	2
pointer	130/154 [77.9, 89.3]	46	16	0	6
cascade	139/154 [84.6, 94]	56	11	0	7
naive, null-allowed	134/154 [80.8, 91.4]	47	11	0	3
trivial: latest date on answer-typed documents	*131/154 [78.6, 89.8]*	*44*	*16*	*0*	*3*

P1: H1 null safety failed (the cascade dates 7 of 90 answerless documents, 6 of them council reports; upper bound 15.2% vs 10%); H2 gave Branch B (cascade alone right on 5, rules alone on 1, p = 0.22), so Apertus left the unattended path for an opt-in review suggestion (D6); H3 held (AUROC 0.969). The 8B cascade: 119/154, 25 of 90 answerless dated.

P2 arm (62 documents)	exact [CI]	answered: found /29 [CI]	wrong date	invented	answerless dated /33
rules only	35/62 [44.1, 68.1]	**2/29** [1.9, 22]	2	0	0
pointer	**53/62** [74.7, 92.2]	**24/29** [65.5, 92.4]	8	0	4
naive, null-allowed, re-parsed (D2-4)	54/62	24/29	7	1	3
trivial: latest printed date	*24/62*	*22/29*	*38*	*0*	*31*

P2: the rules collapse off their home layouts (47/64 → 2/29). H2 held: on all 62 documents the pointer beats the rules on 22 and loses on 4 ($p = 5.3 \times 10^{-4}$); resampling the 15 parliaments, the gain is [-0.02, 0.54], so it sits in a few parliaments. "Answerless" means no qualifying answer date: the pointer's 4 such dates all fall on documents that contain an answer but print no qualifying date; of the 25 with no answer at all it dates none (post-hoc split; across P2 and P6 the 70B closed-set arms dated 0/53 such documents). Against the free prompt with null allowed: ahead on 2, behind on 1.

P3/P3b/P4. Default (metadata + rules) on 29 affairs: type 27/29, answer date 22/27, 27/27 with Apertus suggestions; the addressed body is withdrawn as evidence (every gold value is "executive", D3-3; P3b: C 2/33). P3b replicated by its registered criteria; its answer date (C 20/35) fell below P3's interval, 3 wrong dates from a Luzern rule that overrode OPD's correct date (the product path now keeps OPD's date and flags the record); with Apertus 31/35. P4: 99 of 120 agenda items as virtual affairs, precision 99/99, no decision invented (0/36).

P5. Rules-only pairing: precision 118/143 = 0.83, recall 118/176 = 0.67, *partial* with document-level intervals, *not yet* when parliaments are resampled (precision [0.66, 0.97]); Valais 0/11. H2: rules beat paragraph order 116:5. H4, read as rules vs free generation: 79:11 (the pointer never fired, H3). Recommendation 34/40 by rules; submitter rules 12/40, pointer 24/40, free generation 34/40.

P6 arm (68 documents, 34 answered)	exact	answered: found	answerless dated	printed nowhere
free prompt, null allowed	**62/68**	32/34	4/34	0
cascade v2 (rules, then v2)	60/68	31/34	5/34	0
v2	59/68	30/34	5/34	0
rules only	54/68	20/34	0/34	0
trivial: latest date on answer-typed documents	*51/68*	*25/34*	*8/34*	*0*
model-free cascade (post hoc)	*55/68*			*0*

P6 — reader v2 (an answer-present gate, then the gold guide's decision-before-letter-date rule). No registered hypothesis held: v2 vs free prompting 1:4 (p = 0.38); v2 dates 5/34 answerless documents (upper bound 30.1% vs 15%), all 5 on documents that contain an answer without a qualifying date; v2 vs v1 5:2 (p = 0.45). On all 68 documents cascade v2 beats the rules on 11 and loses on 5 (p = 0.21); the model-free cascade scores 55/68 (cascade v2 ahead on 5, behind on 0, p = 0.06).

P7 — Apertus as verifier of the review queue (`eval/prereg8/`, LOCK8 before the draw; replayed from `data/p7/`). Test frame: 478 eligible review rows outside every audit; 190 blind packets, gold A1 Opus / A2 Sonnet. H1 (primary): promotion precision 120/120, Wilson lower bound 96.9% ≥ 0.90: **holds**. H4: dated precision 95.8% vs 73.2% for a trivial exact-date match, +22.7 points [10.2, 34.3]: **supported**. H3: the rows free generation with a printed-date filter adds are 28/30 right: **not supported**. Base rate: 86.7% of eligible queue rows are answers, but only 40.9% are answers whose own date passes the rule, so the gain is the date check. Post hoc (`results/p7/p7_posthoc_r8.json`): with Apertus's yes/no gate but the adapters' rule choosing among printed dates, 74.9% date-checked; Apertus's own date choice adds +20.9 points [5.2, 33.1], its yes/no gate 1.1; a model-free date match is also right on is-answer (146/146); without Frauenfeld (109 of 214 promotions) 48/48. Applied to the whole queue, 214 of 917 rows go to the tier `promoted_by_apertus`, listed next to the 6,613 counted rows as a pre-sorted batch for review.

What the tests say. Apertus earns its place where layout varies and no rule exists (P2, P3) and as a closed-set verifier with a date check (P7). **Provenance by construction:** no arm that picks from printed dates returned a date printed nowhere, on P1, P2 or P6, and every value has a location; accuracy equals the same Apertus asked freely with null allowed (P2 54 vs 53 of 62; P6 62 vs 59 of 68; P7-H3). **Null safety** is left to the reviewer: the registered null-safety tests on the 70B failed (P1-H1, P2-H3, P6-H2), which is why the default runs no model.

7. Limitations and next steps, reproducibility, declarations

Next steps. Answer dates are tested in depth and Q/A pairs on 40 affairs; tables, annexes and images inside answers are the next extraction target; P4 covers one commune, P6 has no Italian document, no gold set contains federal documents; spans index our own pdftotext/tesseract text (`--opd-text` maps them onto OPD's). With OPD: the Bern, Fedlex and adapter lists and the nightly job on its side, its hash decision, a spot check of the review rows; with the eCH group: the field-by-field meaning test and schema v0.2 (date roles, `QAItem`, five gaps). **Reproducibility.** `make run` (Docker, CPU, clean checkout) recomputes the census and the Bern join, re-derives the P1 rules arm from the source PDFs and the P1/P2 Apertus decisions from the model cache, replays the registered tests from committed files (P7 included), and recomputes every number in `claims.tsv` from its results file, failing if a document prints anything else. **Declarations.** Claude Code agents (Anthropic) wrote most code, documentation and the gold labels; numbers are checked by code. Public AI (may route outside Switzerland) served evaluation only. Report CC-BY-4.0; standards in `docs/mapping.md`.